AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 19 · WORKBOOK

User Need and AI Contribution

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for User Need and AI Contribution before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Retrieval Augmented Generation

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Numbers and Dates

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Multimodal Evidence

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported User Need and AI Contribution trace and record the first step you can explain without help.

Use W2: Trace the worked case for Capstone **requirement**s and corpus. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed User Need and AI Contribution explanation against the supplied facts before discussing it.

Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the User Need and AI Contribution target to a trace and a justified conclusion.

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied User Need and AI Contribution case and show the deciding evidence.

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

Narrow the **need** and **AI contribution**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A student **need**s the exact due date from a course notice. Should the design start by generating a plausible date or locating the notice? Explain the **AI contribution**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 19 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what User Need and AI Contribution adds to Verified Study Buddy and keep the evidence.

Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied User Need and AI Contribution facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied User Need and AI Contribution facts and keep your reasoning.

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or **need**s_review. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied User Need and AI Contribution facts and keep your reasoning.

A report omits this qualification: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied User Need and AI Contribution facts and keep your reasoning.

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied User Need and AI Contribution facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would **need** to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: User Need and AI Contribution

> Define a useful reading helper before choosing its AI features. Explain the labelled step before checking the changed condition.

Define a useful reading helper before choosing its AI features.

A pupil **need**s the meaning of unfamiliar words in an approved glossary. G3 defines habitat as the place an organism lives; no other sources are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/19

User Need and AI Contribution

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Name the need

The pupil wants a glossary meaning, not a made-up example presented as a fact.

2. Choose contribution

The assistant locates the glossary entry and restates it in clear language with G3.

3. Bound the service

An absent word produces "not in this glossary"; a direct glossary search is the baseline.

CHANGE ONE THING

The pupil asks for photosynthesis, which G3 does not contain.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Define a useful reading helper before choosing its AI features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? An absent word produces “not in this glossary”; a direct glossary search is the baseline.

1. Bound the service

2. Name the **need**

3. Choose contribution

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound the service An absent word produces “not in this glossary”; a direct glossary search is the baseline. Read the picture in order.

An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The pupil asks for photosynthesis, which G3 does not contain. Which conclusion is supported by the supplied facts?

1. The assistant reports that the definition is unavailable in its approved glossary.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The assistant reports that the definition is unavailable in its approved glossary. A helpful interface cannot expand its authorised **evidence** by guessing.

Repair a missing building block

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Which part can be corrected immediately from existing **evidence**, and which **need**s follow-up?. Return to this week and explain what you changed.

Review Multimodal **Evidence** (ai-3/13). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

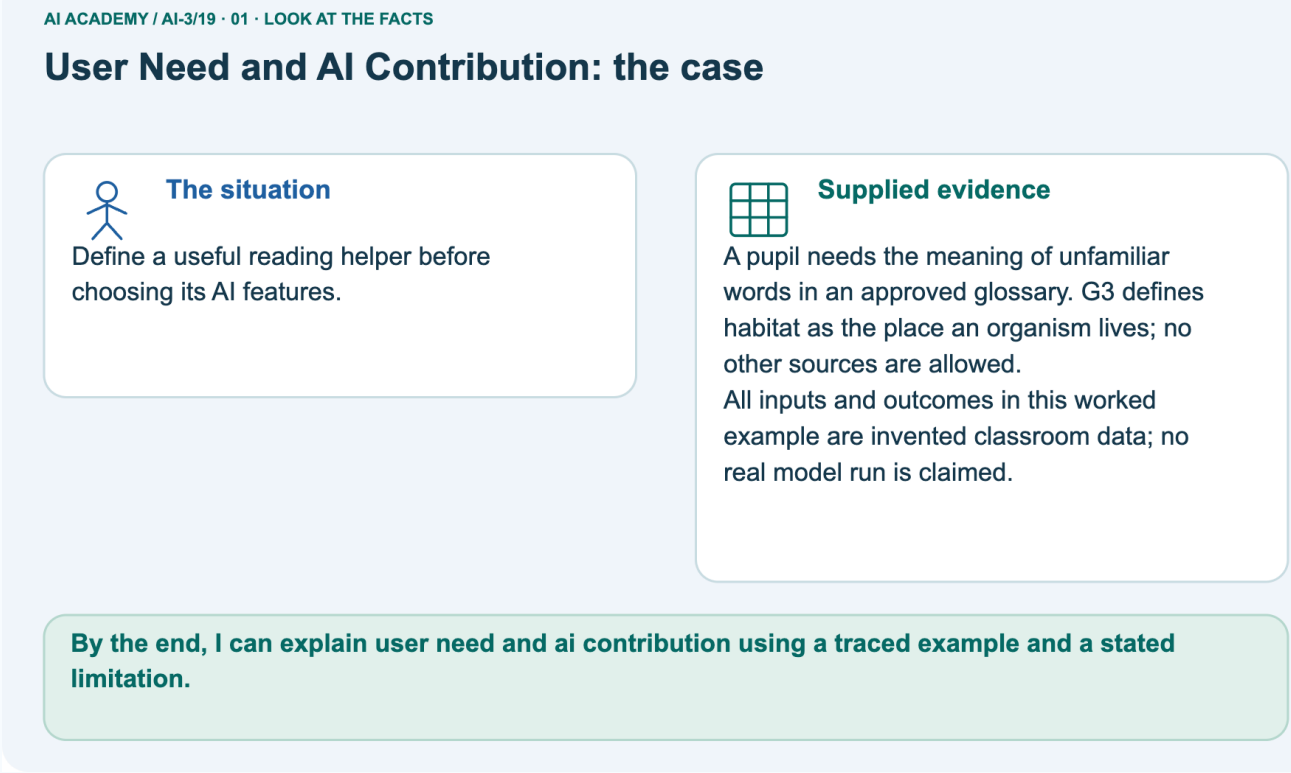
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each User Need and AI Contribution picture from its labels, then record a relevant limit.

01. User Need and AI Contribution: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

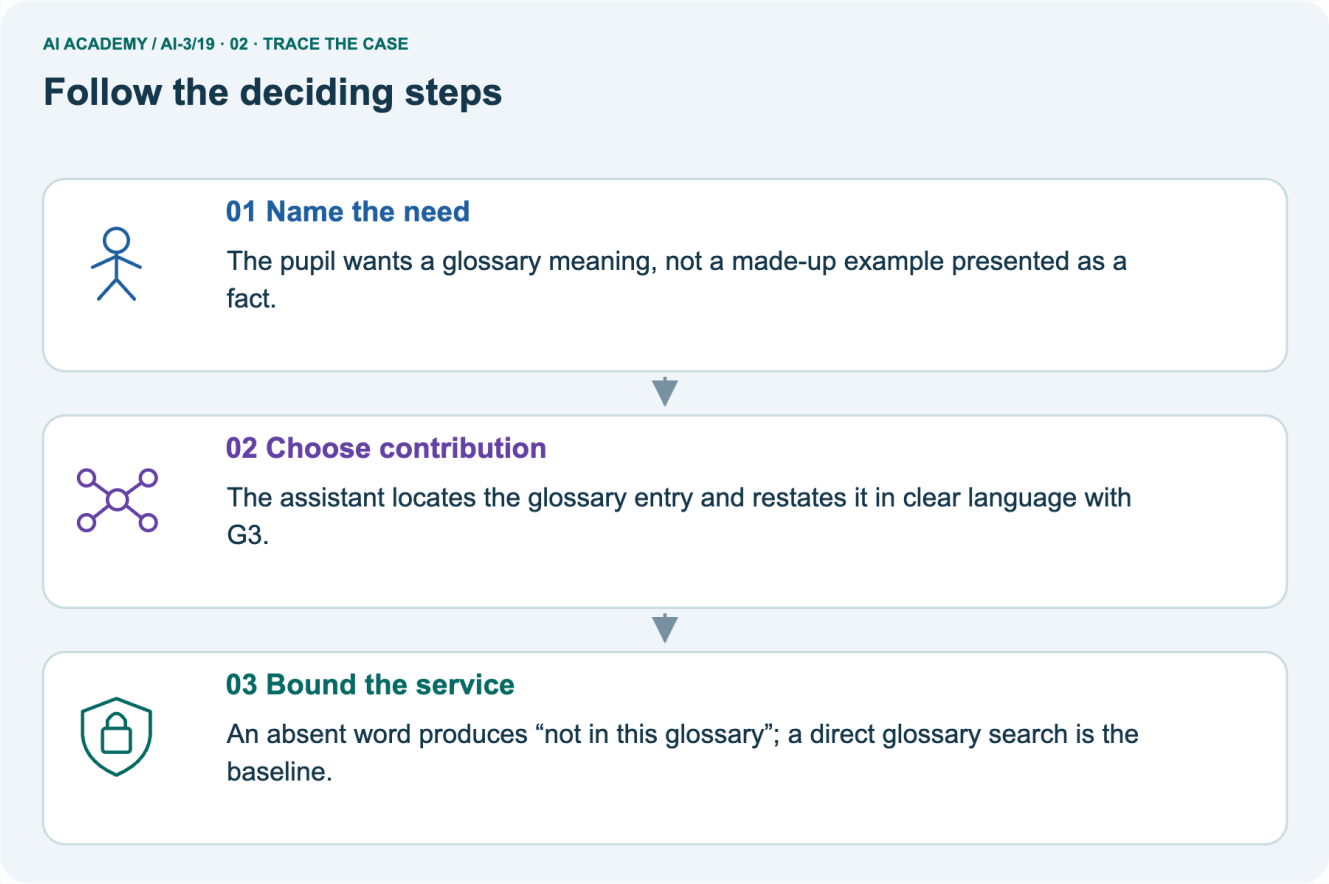
Define a useful reading helper before choosing its AI features. Supplied facts: A pupil **need**s the meaning of unfamiliar words in an approved glossary.

G3 defines habitat as the place an organism lives; no other sources are allowed. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



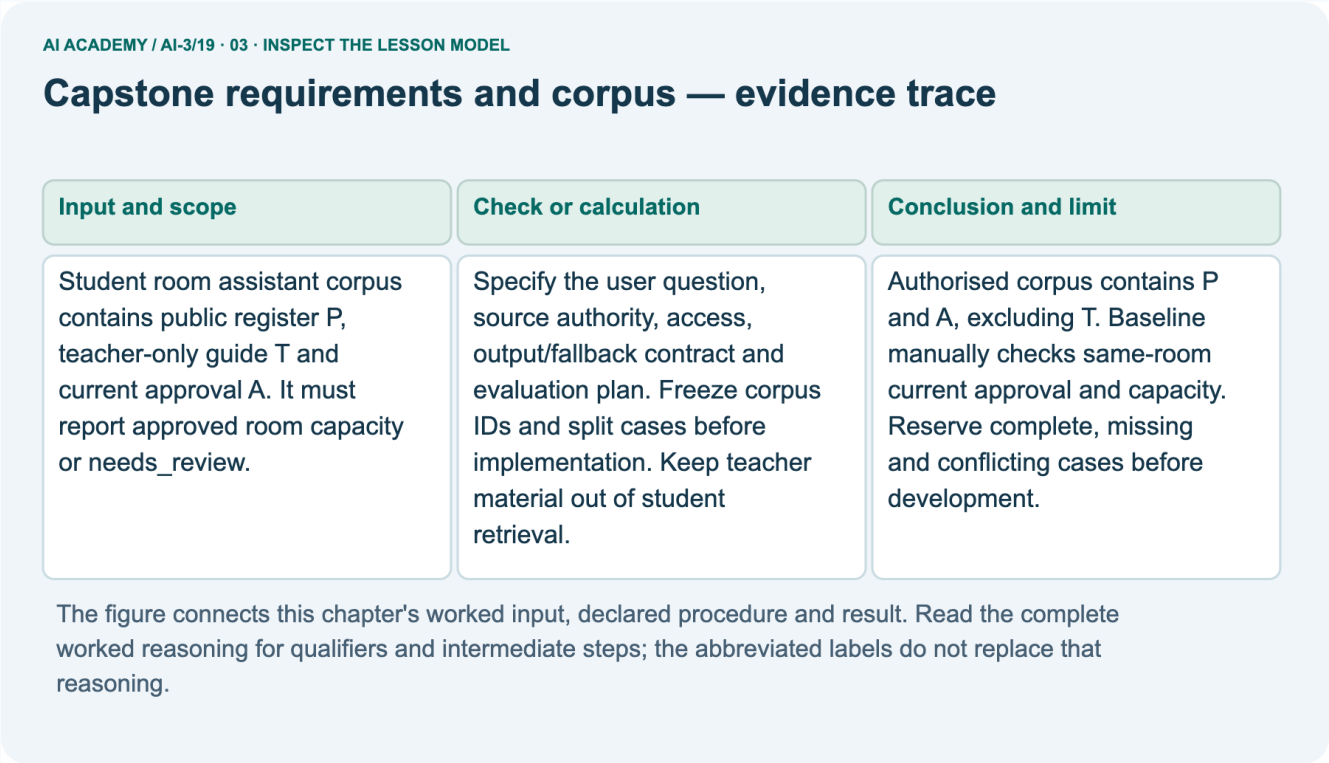
Follow the labelled picture with the matching explanation. Open the vector diagram

Name the **need**: The pupil wants a glossary meaning, not a made-up example presented as a fact. → Choose contribution: The assistant locates the glossary entry and restates it in clear language with G3. → Bound the service: An absent word produces “not in this glossary”; a direct glossary search is the baseline.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone requirements and corpus — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

It must report approved room capacity or **need**s_review. | Specify the **user** question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

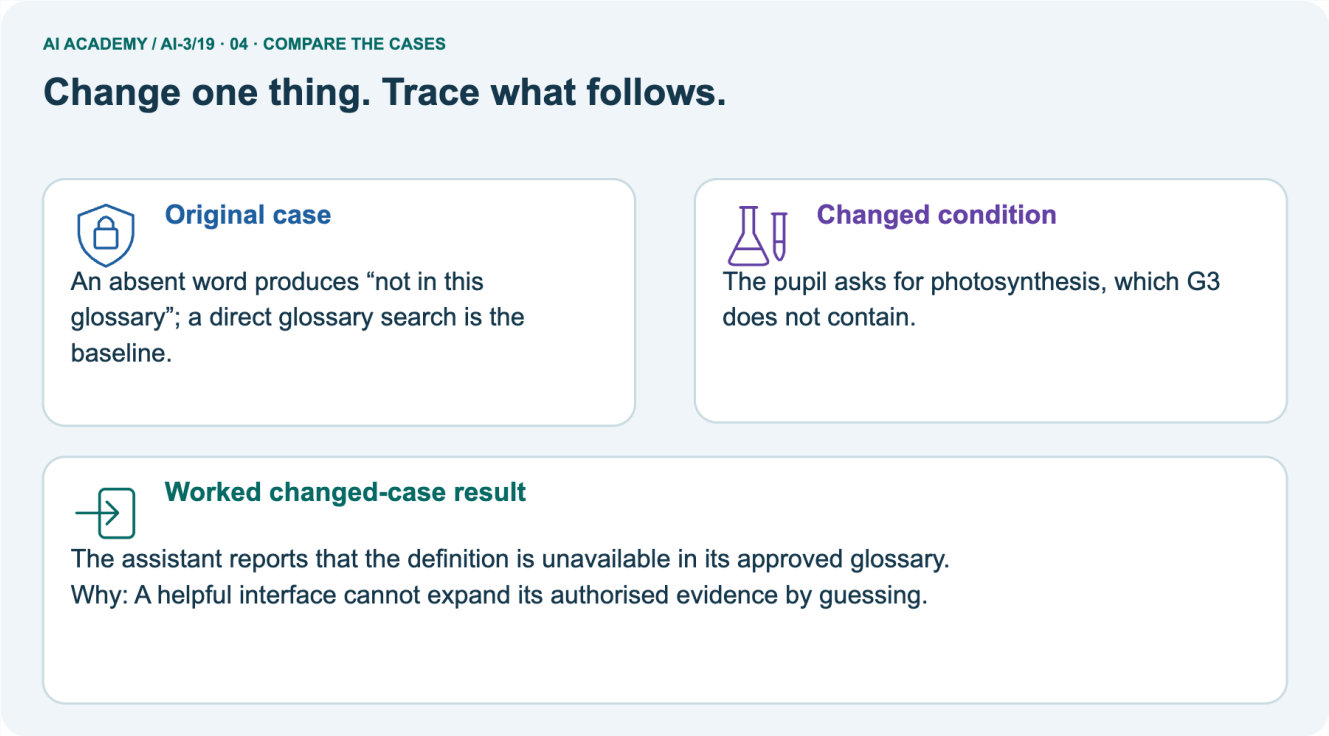
Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

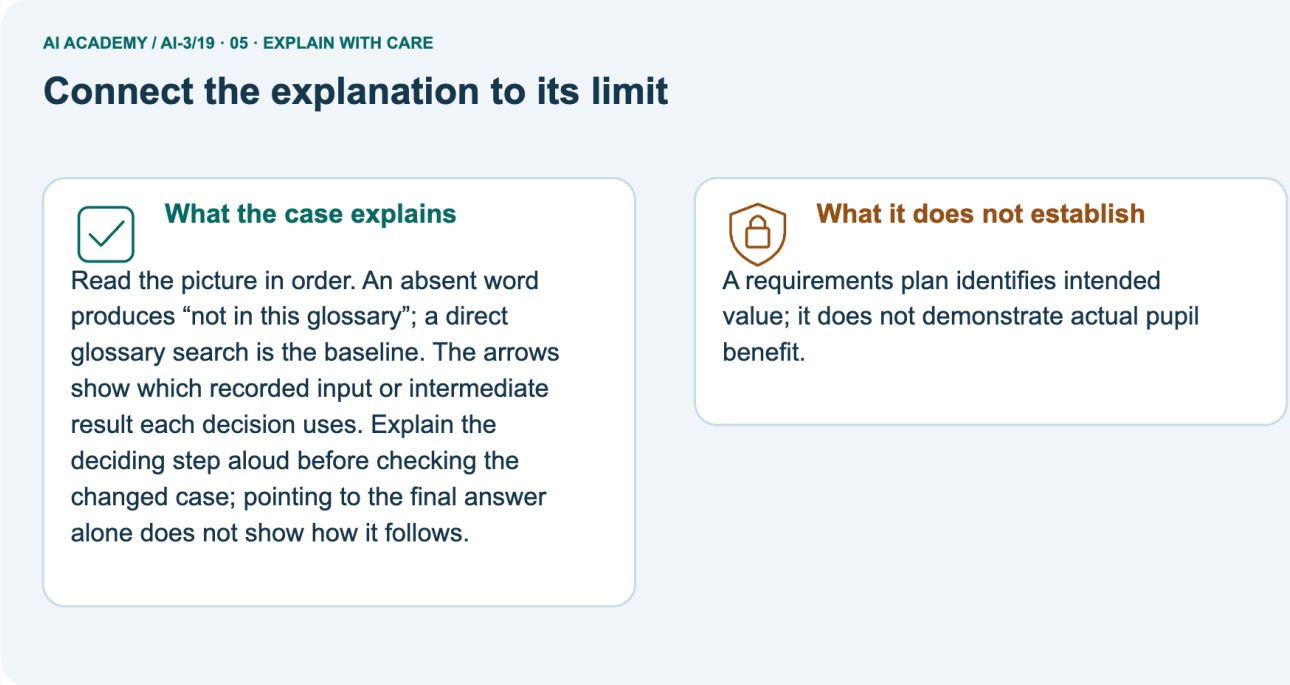
Original case: An absent word produces “not in this glossary”; a direct glossary search is the baseline. Changed condition: The pupil asks for photosynthesis, which G3 does not contain.

Expected result: The assistant reports that the definition is unavailable in its approved glossary. Reason: A helpful interface cannot expand its authorised **evidence** by guessing.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

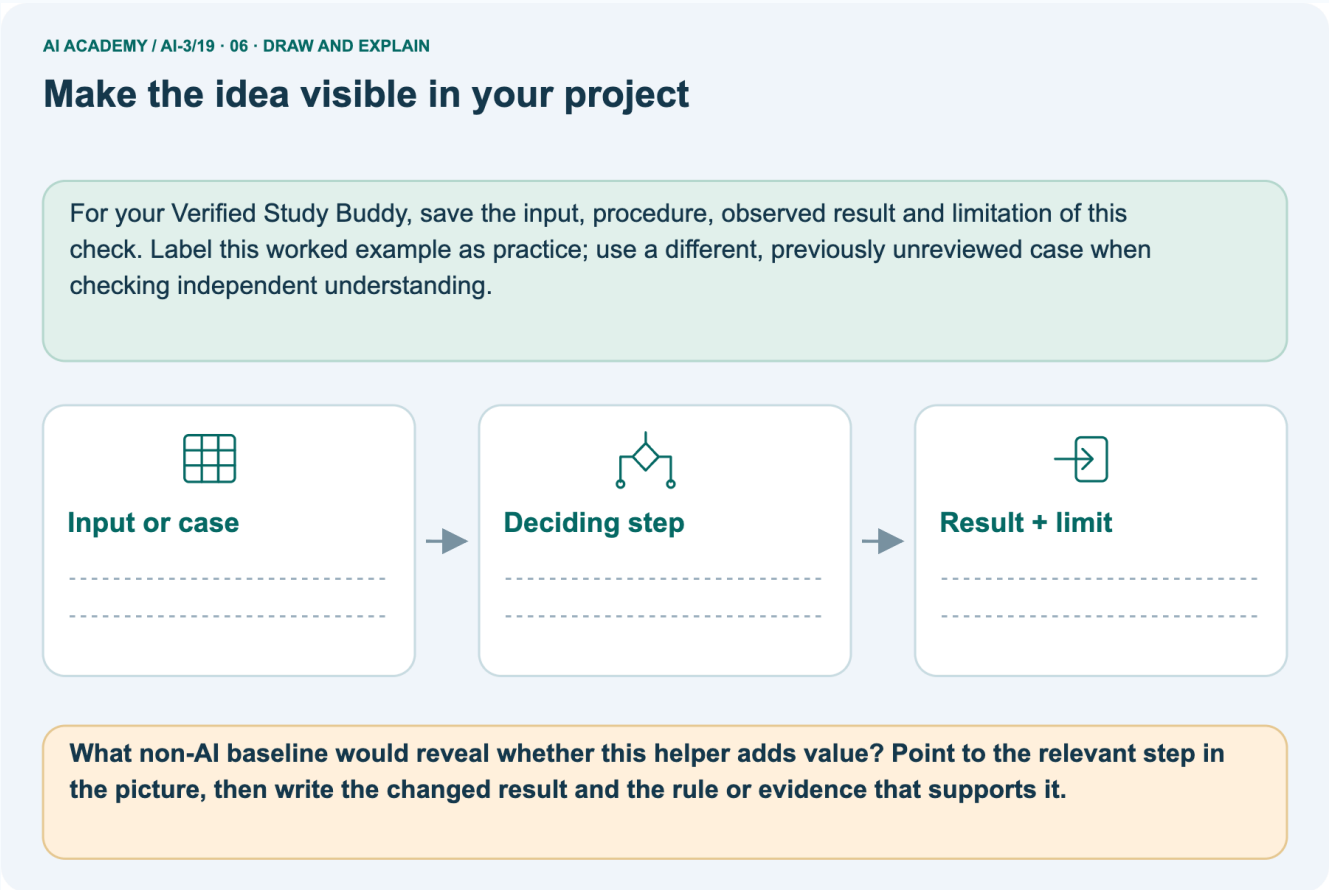
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **requirement**s plan identifies intended value; it does not demonstrate actual pupil benefit.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.